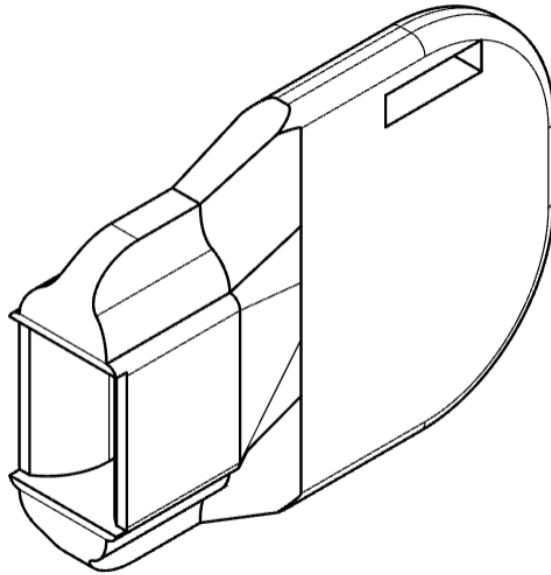


KeyTurn Final Report

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KeyTurn Executive Summary

Lily, a Northwestern student with severe muscle atrophy and loss of sensation in her right hand, struggles with poor grip strength. Everyday tasks, such as turning the thumb-turn locks on her doors, are challenging. She resorts to using her left hand, which is inconvenient due to the right-swing doors in her house. This becomes even more problematic when carrying items like laundry or garbage bags.

We present the KeyTurn, a transportable device used to unlock any thumb-turn locks the user encounters. The main requirements are to increase the surface area that our client could hold and maintain a secure connection between the KeyTurn and the thumb-turn lock during use. These requirements were defined by user testing of the device to determine whether the KeyTurn has sufficient surface area and connection strength to easily turn the thumb-turn locks in her house.

The KeyTurn, is a small, key-like 3D-printed device that fits around the thumb-turn, allowing the user to unlock the door while greatly reducing the amount of grip strength needed. The KeyTurn consists of two main parts: a socket that fits tightly around the thumb-turn and a handle that allows the user to grip and turn the KeyTurn. The socket and the handle are printed as a single device since this method creates the most secure attachment between the two components.

The socket has a rectangular hole with a fillet. The dimensions of the original hole match the size of the thumb-turn and have a fillet with a radius that accommodates the curve near the bottom of the thumb-turn. This shape allows the socket to fit onto the thumb-turn lock securely. The handle is designed to increase the surface area of the device, allowing the user to operate it without needing to exert much force. The size of the handle was inspired by a common item such as an AirPods case since the device should be portable and fit in the user's pocket. The back of the handle is rounded at the top and bottom to reduce the device's overall footprint while retaining its ergonomic shape.

One application of the KeyTurn involves using a retraction mechanism. The retraction mechanism is a retractable ID holder that can clip onto a hole at the top of the KeyTurn handle. This mechanism allows users to clip the device onto a backpack, belt loop, or other article of clothing, keeping it easily accessible.

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Introduction

Our project partner, Lily, is a Northwestern student with severe muscle atrophy and loss of sensation in her right dominant hand [1], and struggles with poor grip strength. Since she can't grip small objects, everyday tasks such as turning the thumb-turn locks on her doors (Figure 2) are challenging. She resorts to unlocking doors using her left hand, which is inconvenient due to the right-swing doors in her house. This becomes even more problematic when carrying items like laundry or garbage bags.

Her current method of using her left hand to unlock and open doors is inconvenient since the doors in her house swing open from left to right, making them more comfortable to operate right-handed. Difficulty using her left hand also becomes apparent when she needs to operate her doors while carrying other objects, such as laundry or garbage bags.

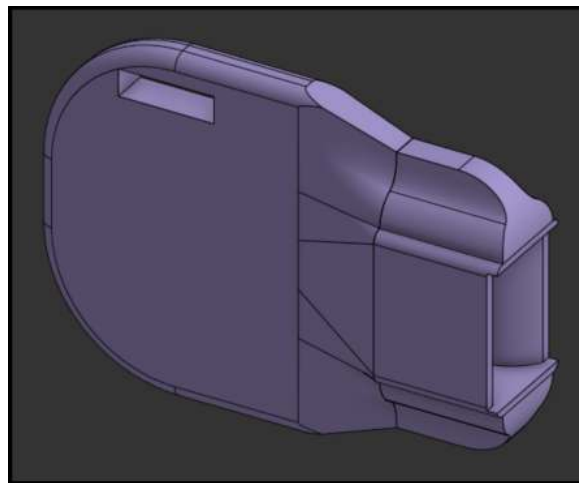


Figure 1: KeyTurn Device

We present the KeyTurn (Figure 1), a transportable device used to unlock any thumb-turn locks the user encounters. This device is shaped like a key, and fits around the thumb-turn, allowing Lily to unlock her door as if she's using a key. This solution allows Lily to turn the

thumb-turn lock on her doorknob with her right hand, making her experience of locking and unlocking doors more efficient. Our solution has two main parts: a socket that fits tightly around the thumb-turn and a handle that allows the user to grip and turn the lock comfortably. By using a device with a larger surface area that fits around the thumb-turn, Lily can unlock her doors more easily without needing to exert much force. This report indicates how the user should interact with the design as well as providing evidence and rationale towards guidelines for usage.

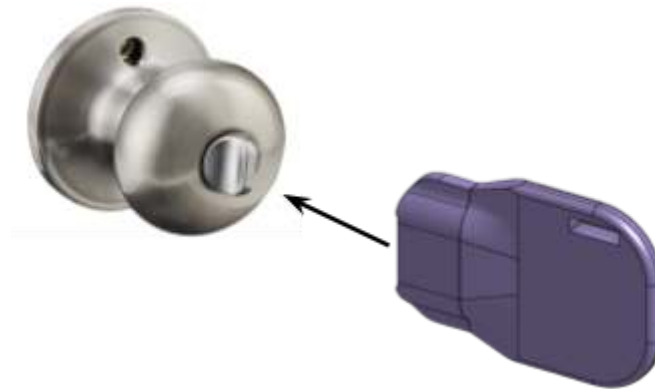


Figure 2: Thumb-turn style door knob with KeyTurn

Users and Requirements

Our project partner notes difficulty operating small objects that require fine motor skills with her right hand. Notably, she has difficulty operating her house's thumb-turn door locks. While Lily is the primary user in the main focus of the design, secondary users include people with poor grip strength in the combination of their thumb and any other opposed finger or people with a similar condition to Lily.

Our design has several requirements: 1. The device must successfully fit over and rotate a thumb-turn door lock. 2. It must reduce the force required to turn a thumb-turn. 3. It must be compatible with a variety of common thumb-turn door locks, especially the ones in Lily's home.

4. The device must be ergonomic and easy to use, catered towards individuals with weak grip strength, providing stability and ease of manipulation. 5. The device should be durable and withstand repeated use over a long period of time. 6. The device's dimensions should be small enough to fit in a pocket, similar to commonly used objects such as Apple AirPods.

Design Concept and Rationale

The KeyTurn (Figure 3) is a compact 3D printed device to help individuals with weak grip unlock doors with small thumb-turn locks. The KeyTurn consists of two main components: The handle and the socket. The handle provides a secure grip on the KeyTurn . The socket fits tightly around the thumb-turn lock itself, allowing the user to turn the device like a key to unlock the door . Combining these two components allows users to easily unlock doors with thumb-turn locks. A modular version, where the head of the KeyTurn can be swapped out for different door knob sizes and shapes, could make the device more adaptable for various environments. Although prototypes for modularity were explored and tested, these concepts were never furthered due to concerns over complexity.

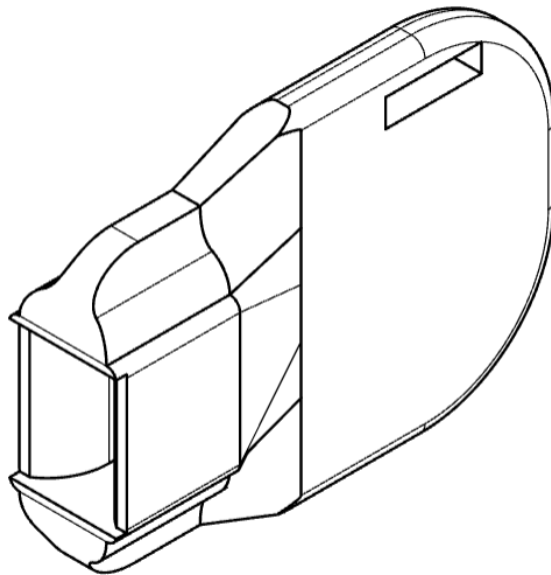


Figure 3: Isometric view of the KeyTurn

Top View



Front View



Side View

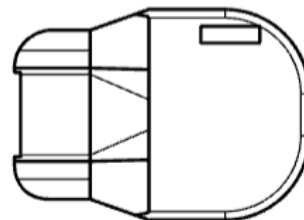


Figure 4: The front, top and side view of the KeyTurn

The socket and the handle are printed as a single device since this method of creating the device creates the most secure attachment between the two components. Printing the components separately was considered for previous iterations as this method may allow the device to have a higher print quality; however, the junction between the two components was less secure than if the entire object was printed as a whole. Additionally, printing these components together simplifies the manufacturing process, allowing the device to be produced in a shorter timeframe and with fewer materials.

The retraction mechanism attaches to the KeyTurn, allowing the user to easily carry the KeyTurn (Figure 9). The following sections describe the components of the device: the handle, the socket and the retraction mechanism, along with the rationale behind their design.

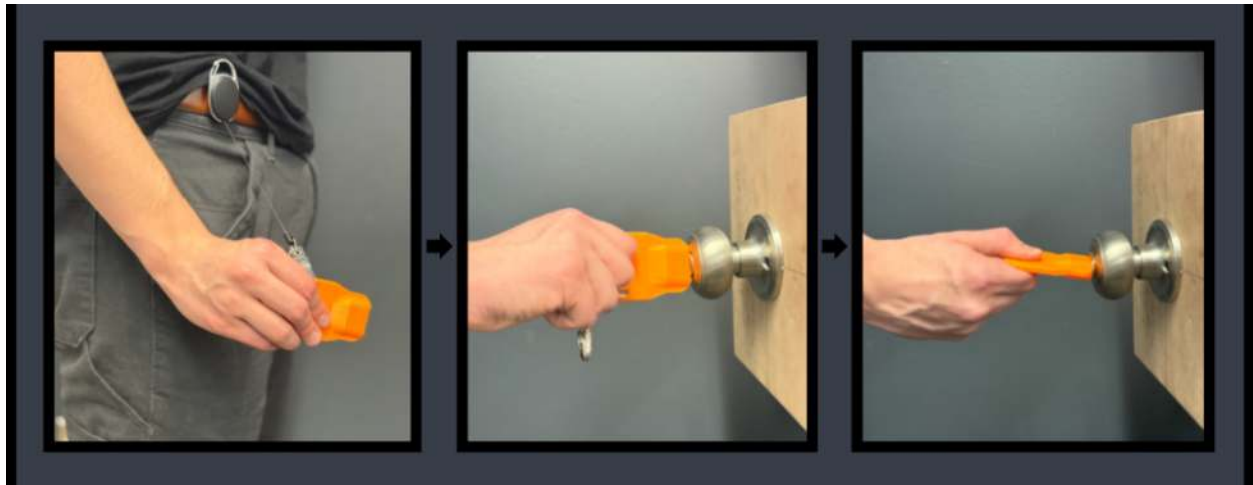


Figure 5: Demonstration of using the KeyTurn in 3 steps: (1) pulling the KeyTurn from the waist, (2) inserting the device onto the thumb turn lock and (3) turning the device.

The Handle

The handle allows the user to securely grip the KeyTurn and turn the lock on the door. The handle essentially increases the surface of the handle, allowing the user to turn the lock without needing to exert much force (Figure 6).

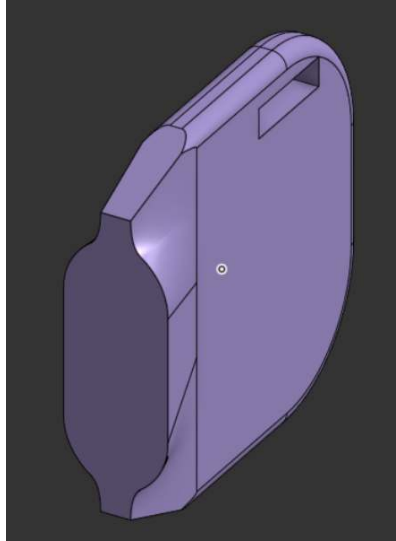


Figure 6: 3D model of the Handle of the KeyTurn

The key style grip provides power and precision for people with arthritis [2]. The width and shape make it portable while ensuring a firm grasp, enabling one-handed operation. The edges are rounded for comfort and safety, The handle also includes an attachment hole for securing the device to the retraction mechanism or other carrying options. The handle's height and length were determined by measuring the dimensions of the average womens' pants pocket size so the device will fit in a small pocket [4]. The Handle is also flat so that it will sit flush in a pocket without any major protrusions. The handle is made of Polylactic acid plastic (PLA), which is lightweight but strong [5], ensuring the device remains sturdy while being easy to carry. A rectangular hole is integrated into the handle to accommodate the retraction mechanism, allowing attachment to keychains, belt loops, or bags for convenience.

The Socket

The socket is the portion of the device that attaches to the thumb-turn lock on the doorknob, allowing the user to turn the lock (Figure 7).



Figure 7: 3D model of Socket of KeyTurn

The dimensions of the socket are 18 x 40 x 14 (LxWxH) mm. The top and bottom of the socket include fin-shaped supports because they increase the surface area between the socket and the handle. The front of the socket has a rectangular hole with a fillet. The dimensions of the original hole match the size of the top edge of the doorknob, 20 mm (height) by 4 mm (width), with a depth of 18 mm. This hole has a fillet with a radius of 7.5 mm to accommodate the curve near the bottom of the doorknob, allowing the socket to fit securely onto a variety of thumb-turn locks. The socket is made of PLA like the handle since this material can be easily printed to the desired shape while also being durable enough to withstand the forces during normal use.

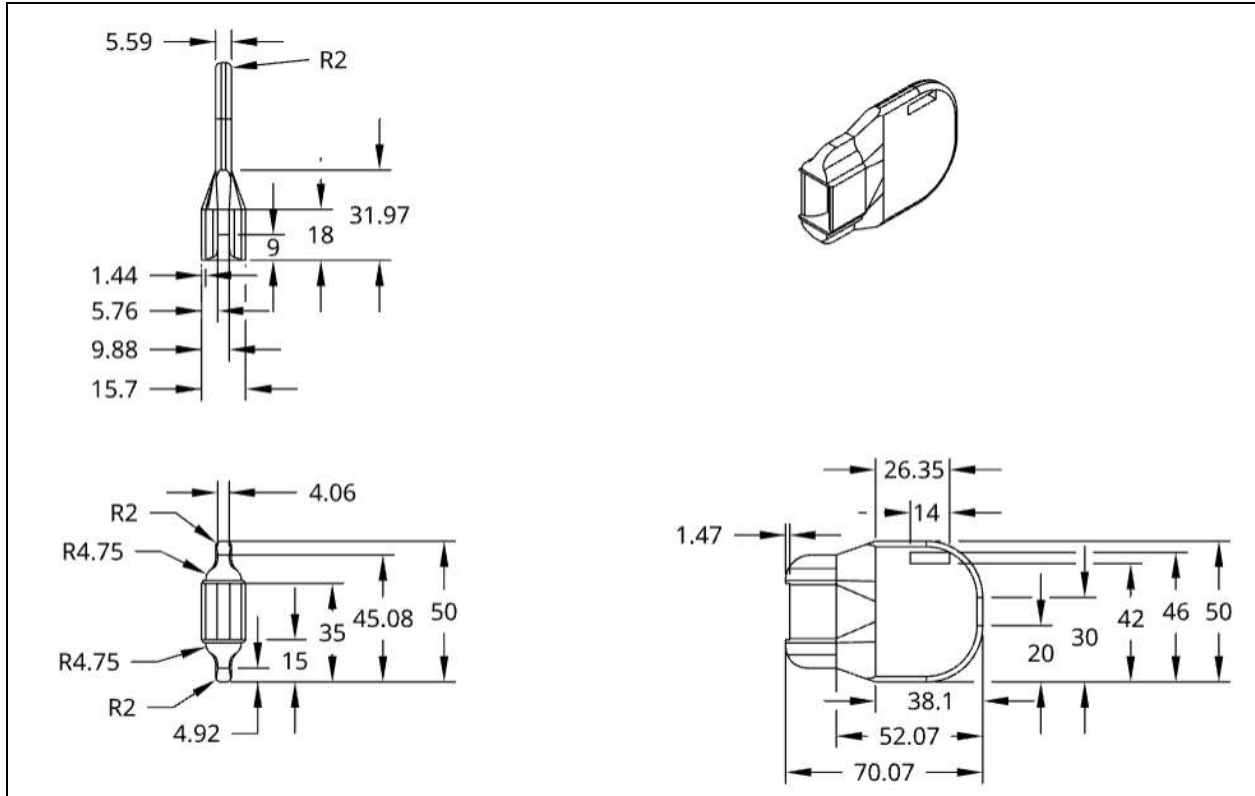


Figure 8: Orthographic view of the KeyTurn composed of the Socket and Handle (in mm)

The Retraction Mechanism

The retraction mechanism is a retractable ID holder that can clip onto a hole at the top of the KeyTurn handle (Figure 9). This mechanism allows users to clip the device onto a backpack, belt loop or other article of clothing, keeping it easily accessible without needing to store it in a pocket. This allows the user to easily operate the device using a single hand and eliminates the need to put away the KeyTurn as it can retract back into position by simply letting go.



Figure 9: The retraction mechanism that attaches to the KeyTurn

Future Developments

The current iteration of the KeyTurn succeeds in allowing Lily to operate thumb-turn locks with her right hand, but further refinements can improve its accessibility for a wider range of users. While the device effectively addresses Lily's needs, there is a much broader audience of people with grip strength issues similar to Lily, such as the elderly or people with similar symptoms such as cerebral palsy. With further testing and research, we can see if the KeyTurn is a viable solution for this broader audience, and if not, what changes need to be made to the Keyturn to make it more accessible?

The shape of the KeyTurn could be adapted or made interchangeable to accommodate different door hardware. Beyond improving this specific device, collaboration with organizations such as Design for America or integration into the Northwestern University "Design Thinking and Communication" course could further its impact.

Conclusion

The Keyturn successfully addresses the challenges Lily faces in unlocking thumb-turn style locks, providing her with a solution that is both practical and ergonomic.

The KeyTurn features a secure socket that fits tightly around the thumb-turn lock, preventing slippage during operation. The handle is shaped to optimize grip strength using a key-style grip, which balances power and precision while being particularly suitable for users with poor grip strength. This ensures that Lily can easily and effectively turn the lock with minimal force, meeting the requirement of reducing strain and effort. Portability is another aspect of the design that provides practicality. The small dimensions allow for it to comfortably fit in

the pocket without creating a noticeable protrusion. This combined with the badge reel work together to maximize ease-of-use.

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Appendix

Project Definition

Mission Statement

Lily has difficulty turning thumb turn locks in her house due to limited strength or mobility in her dominant hand. Design a portable physical device that allows/provides enhanced grip and force application, thus reducing the time and effort required for her to unlock doors.

Design Constraints

- The doors cant be modified.
- Low cost of production due to \$150 budget
- Limited to 10 weeks
- It must follow safety regulations: avoiding sharp edges or harmful materials.

Users and Stakeholders

Lily L., user and project partner, will use the device to restore her ability to open thumb turn doors with her dominant hand.

Lily's roommates live with her off-campus, and could encounter the use of this product depending on the final design.

Anyone with limitations in hand dexterity.

User Profile

Lily is a Junior at Northwestern University who has developed muscle atrophy in her right (dominant hand). She has difficulty in engaging with objects that require finer movements such as thumb-turn door knobs so she uses her left hand instead. However, this is not her dominant hand, and she would prefer to be able to utilize both hands.

Illustrative User Scenario

Lily regularly encounters challenges whilst operating the thumb turn on numerous doorknobs in her apartment. She deals with a pair of doors while exiting the apartment. The first door is a left-handed door which presents less of a challenge given that the door is designed to be turned with her left hand, and also has a handle rather than a knob. Lily finds greatest difficulty in operating the door to exit the apartment as the door is not only right-hand operated, but also has a rounded thumb-turn lock. Once she steps out of the apartment, she must reach in to lock the door from the inside which is a challenge while using her left hand. Lily also struggles with a door towards the rear of her apartment leading to the basement laundry room. She must unlock the door with her left hand whilst holding her laundry, then swap the laundry into her right hand whilst turning the door and exiting down the stairs.

Table 1: Needs Identification, Metrics, and Specifications for prototype

Doorknob Device Specs <i>Requirements in Metrics Version 3.0</i>					
Requirements in Metrics	Metrics	Units	Ideal Value	Allowable Value (Estimate)	As-Built
Easily transportable	Weight	g	<=100 Based off of average comfortable weight for cell phone	<=150	0.056 lbs ~ 25.4 grams
	Max length	mm	<= 120 mm based off of the average women's pocket width	<=180 mm	70 mm
	Max width	mm	<= 8 mm based off of average comfortable depth of cell phone	<= 15 mm	14 mm
	Max height	mm	<= 95 mm based on the average women's pocket depth	<= 185 mm	50 mm
Safe for user	Edge Fillet	mm	>= 2 mm Based off of user's preference	1 mm per the guidelines for the minimum fillet size	2 mm

User Observation

Introduction: This report summarizes findings and observations from Lily, our client, who has mobility challenges in her right hand due to a condition known as Chiari malformation. The purpose of this study is to understand the specific difficulties she faces in her day-to-day life, particularly with the operation of thumb-turn locks and doorknobs in her apartment. The insights gathered here will help devise an assistive device design aimed at mitigating such challenges.

Methodology: The study combined interviews and observations.

- Interview: Conducted in class with Lily to gather insights regarding her condition, her day-to-day struggles, and preferences in potential solutions.
- Observation: Lily demonstrated challenges with various doors in her home, along with photos and measurements.
- Data collection: Individual research, findings, notes, and recorded demonstrations were used.

Results: The following outline key findings from the interview and various observations:

Table 1: Observations and User Interpretation of Hand Dexterity

What we saw	What was said about it
Struggles in twisting the thumb-turn lock on the right-handed door leading to the front	“This one is a bit harder, because when I use the door I pre-lock it, but because it is a

porch	right-handed door, I have to block it with my foot and it's hard for me to turn.”
Uses the bannister railing on the right side of the steep staircase after the basement door.	“These stairs are very steep, so if I’m holding my laundry on my right hand and using my left hand to turn the door, I have to swap hands to hold the railing which is very inconvenient.”
Demonstrated challenges with gripping small objects like a fine thumb turn.	“The issue is the grip and the turning together. Larger objects are easier for me to hold.”
Struggled to twist small bottle caps or thumb turns but could manage larger, wider caps.	“I don’t like little little bottle caps, at that point I’ll literally just ask someone for help or use my teeth.”

Additional Observations:

1. Grip Strength and Dexterity:

- a. Lily has limited grip strength and dexterity in her right hand but retains mobility in her wrist and thumb. She struggles with fine motor tasks and relies heavily on visual feedback due to the lack of sensation in her right hand.
- b. She prefers to use larger objects as they allow her to use her entire hand rather than specific fingers.

2. Environmental Context:

- a. The apartment doors are all different and require slightly different approaches due to their handedness and lock types.
 - b. Carrying items like laundry exacerbates difficulty as she needs her left hand to carry those heavy objects.
3. Existing solutions and preferences:
- a. She hasn't encountered any specific assistive devices tailored to the thumb-turn locks. Most existing solutions are either too broad or not applicable.
 - b. Prefers to have a portable solution that can be easily carried or stored.
 - c. Wants to avoid solutions which might inconvenience her roommates or require excessive time.

Photos:



Figure 10: Door leading to her porch



Figure 11: Door leading to Lily's bathroom



Figure 12: Lily is seen operating the door leading to her basement laundry room. While she can operate the door itself with ease, she is often holding a heavy laundry basket which she must exchange into her left hand afterwards to use the bannister railing.

Discussion: Considering Lily's feedback and observations, the following considerations should guide the design of an assistive device:

- Key features:
 - Versatility: The device should work on various thumb turn locks and knobs, including those in Lily's apartment and other locations.
 - Ease of Use: Minimize steps to use the device, ensuring it does not add unnecessary time or effort to daily tasks.

- Enhanced Grip and Surface Area: Incorporate a design that increases the surface area for gripping and allows engagement of the entire hand.
- Portability: Ensure the device is lightweight and compact enough to fit in a pocket or bag.
- Material and Design:
 - Use non slippery materials to enhance grip such as silicone
 - Avoid complex mechanisms that could burden Lily or her roommates.
- Additional insights:
 - Device should be able to accommodate varying thumb turns.

Conclusion:

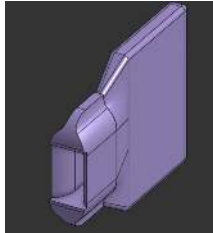
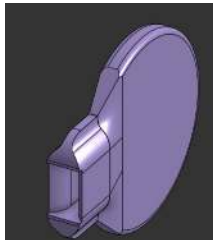
- The insights gathered from this study emphasize the importance of designing a practical, user-friendly assistive device tailored to Lily's specific needs. The device should prioritize ease of use, portability, and adaptability to ensure it integrates seamlessly into her daily routine while addressing her challenges with grip strength and dexterity.

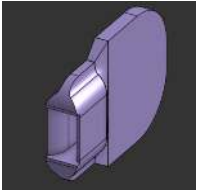
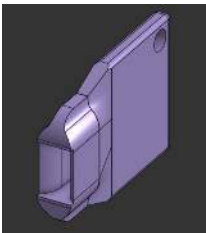
User Testing Results

Introduction

Our team visited Lily at her house to test different versions of the KeyTurn. The following table illustrates her preferences and feedback to each of the following models: KeyTurn v2.0 Circular, v2.0 Rectangular, v2.1 and v2.2.

Table 1: User Testing results and the respective tested model

Name	Changes	Model	Feedback
KeyTurn v2.0 Rectangular Handle 2/12/2025	- Added supports on top and bottom of device - Removed holes for magnets - Added loft to transition from headpiece to handle - Reduced handle size - Added fillet to inside of headpiece to accommodate the doorknob lock		- Likes the overall shape of the design - Shape of the handle is slightly off, our team may want to increase the fillet on the top and the bottom of the back edge of the handle. - KeyTurn fits very well to her front door
KeyTurn v2.0 Circular Handle 2/12/2025	- Added supports on top and bottom of device - Removed holes for		- Thinks the handle is too large and some of the space on the top and bottom are

	- magnets - Added loft to transition from headpiece to handle - Greatly reduced handle size - Added fillet to inside of headpiece to accommodate the doorknob lock		unnecessary for normal operation - Socket of the KeyTurn still fits very well
KeyTurn v2.1 2/13/2025	- Removed 2 piece print and combined to form single print - Reduced handle height		- Likes the shape of the handle much better compared to the v2.0 designs - Socket of Keyturn is has a very snug fit on her doorknob - Silicone on tip of KeyTurn assisted the grip of the socket to the doorknob
Keyturn v2.2 2/18/2025	- Slightly increased width of handle and support braces on top and bottom of headpiece - Added hole to allow		- Lily much prefers the retraction mechanism to clip the device onto her waist - Prefers the handle shape of the KeyTurn v2.1 compared

	keychain attachment - Increased distance of loft		to this rectangular shape
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Conclusion

Overall, the design fit very well onto her doorknob. The KeyTurn v2.1 has silicone attached to the tip which greatly assists the KeyTurn socket in attaching to the doorknob. Additionally, the user greatly prefers the attachment mechanism present in v2.2 compared to putting the device in her pocket for the other designs. Attaching the KeyTurn to the retraction mechanism may present some difficulties, so a rectangular hole may be used to better accommodate the clip on the retraction mechanism. The user describes the surface area and dimensions of the KeyTurn v2.1 as being almost perfect for her average use case, since it is large enough to hold onto while also small enough to not interfere with her daily activities.

Performance Testing

Table 1: Performance tests completed on the KeyTurn and their results

Test	Description	Result
Drop Test	The KeyTurn was dropped 100 times from approximately waist height (~2 ft from the ground). Damages were measured through visual inspection after each trial.	No significant deformities or damages to the KeyTurn after testing
Retraction Test	The KeyTurn was attached to the retraction mechanism. The KeyTurn was pulled out to its maximum length and snapped back into place 50 times. Damages were measured through visual inspection and feedback from tester	No significant damage to the user, the retraction mechanism or the KeyTurn after testing
Unlock Test	The KeyTurn was attached to the 4 members of our DTC design group and each of us used the KeyTurn 50 times on a doorknob set up on a table. The accuracy of the test was recorded by measuring the number of times the user could unlock the door divided by the number of total times the device was used.	Kevin: 48/50 (96% accuracy) Lucas: 49/50 (98% accuracy) Daniel: 47/50 (94% accuracy) Mason: 49/50 (98% accuracy) On average, the accuracy of the KeyTurn was 96.5% for across all testing

Background Research

Section 1: Muscle Atrophy & Related Diseases

Chiari Type I and Muscle Atrophy

Our project partner, Lily, is a young adult woman diagnosed with Chiari malformation. Chiari Type I is a disease caused by leaking of the spinal fluid from the lower back or chest regions of the spine and is believed to cause muscle atrophy and a loss of sensation in her right dominant hand. According to John Hopkins Medicine, some cases of Chiari Type I can affect neuromuscular function, leading to weakness in the limbs [1].

Our client's muscle atrophy is more prevalent in her ring and pinky fingers. According to Hemani Ticku and Bashar Katirji, “Atrophy of interosseous and hypothenar muscles with sparing of the thenar muscles is often due to ulnar neuropathy at the elbow or wrist” [2]. The interosseous and hypothenar muscles are the muscles that control the ring and pinky fingers, which likely suggests that the Chiari Type I has affected the ulnar nerve in the elbow through an expansion of fluid in her spine.

Related Diseases

There are numerous diseases that also cause ulnar neuropathy, the damage of the ulnar nerve. One is named Ulnar Nerve Entrapment, a condition that occurs when the patient has sustained severe damage to their elbow [3]. In some personal anecdotes, patients reported experiencing extreme numbness in their pinky and ring finger, similar to our client. In these cases, the patient would usually be fitted with a cast for a few days until they can get the necessary medical procedure to fix the issue [4]. Another similar condition is cubital tunnel syndrome, where injury or other related cause causes damage to the elbow, causing pain and

numbness in the arm or hand. In Jasmine Nakkab's case, this caused numbness and pain in her left ring and pinky finger, similar to the symptoms of our client [5]. The purpose of researching similar diseases is to examine considerations and solutions to these conditions that could be implemented into our design. Samir Sharrak and Joe Das give good insight on this topic, as they write "Initially, it is important to instruct the patient to avoid repetitive use of the affected extremity. Wearing wrist splints is typically recommended during night-time or symptom-provoking activity" [6]. Designing a product around these types of conditions ideally should reduce repetitive use of the affected muscles. In addition, hand braces may be beneficial to use when performing any activity that uses the affected muscles.

Biomechanics of grip strength

Grip strength is a complex biomechanical phenomenon primarily determined by the force generated by the forearm flexors and the intrinsic hand muscles. These muscle groups are critical for generating the power required to hold and use objects effectively.

The primary biomechanical contributors to grip force are the joint angles and torque at the wrist and finger joints. Proper hand and wrist positioning is critical for measuring grip strength consistently. R. W. Bohannon highlights that grip strength serves as an indispensable biomarker for overall health, particularly in older adults [7].

While muscle and grip strength are vital, friction between the hand and the gripped material can significantly impact the ability to grip adequately. For instance, materials with higher friction coefficients, like silicone, tend to provide better grip stability. Studies from a collaborative group demonstrated that silicone has the highest friction coefficient with human skin, recorded at 0.61 with a standard deviation of ± 0.21 .

Functional grip for medical assistive devices must allow for the full range of finger flexion movement. The ultimate design goal is to enable, not restrict, natural hand function.

As noted by K. Balogun et al. [8], hand and wrist positioning during grip strength assessments ensures reliability and repeatability. Testing done with various wrist rotation degrees can lead to biased results and may even lead to critical failure during actual use.

Section 2: Prototype Design

Materials with high frictional Resistance

The three main types of friction materials are organic, inorganic and metallic. Organic and metallic materials are the two types of materials relevant to this project. There are two main types of organic materials, those based on the mineral asbestos and those not. Organic materials such as low steel are used in car breaks because of their high frictional properties [9]. Metallic materials such as copper, brass or steel are also commonly used in car breaks for their high friction [9]. Another material with high frictional resistance is rubber such as butyl rubber, styrene-butadiene rubber (SBR) and silicone. It may be more beneficial to use rubber for this project because it is easier to mold than metal. Silicone is a good option for prototyping because out of all the rubbers it generates the most friction when in contact with human skin [10].

Moldable Materials

There are two main methods that can be used to print the prototypes of our design: Injection molding and 3D printing. Injection molding is able to create a high quality version of our design, since we can use materials such as Acrylonitrile Butadiene Styrene (ABS) and High-Density Polyethylene (HDPE) [11]. Although these materials are exceptionally resilient,

creating the mold and purchasing a Injection molding device typically requires thousands of dollars and our shop does not have an injection molding machine.

The other method of creating our prototype is through 3D printing. This is a far more accessible method of creating prototypes, especially since our shop has a large number of 3D printers for students to use. The overall print quality mainly depends on the 3D printer, the type of filament used and the temperature at which it prints. Some of the high quality filaments that are commonly used are: ABS filament, which has a high durability compared to other filaments, and PLA, which is easier to mold but less durable [12].

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Instructions for Creation

Tools Needed:

- 3D printer
- Computer capable of running modern 3D printing software

Materials required:

- PLA 3D printing filament

Section 1: Bambu Studio setup

1. Download file

Download the stl file for [KeyTurn 3D model](#)

2. Import Model:

Open Bambu Studio and import the downloaded .stl file

3. Select Printer and Plate Profile

Under **Prepare > Printer**

- select Bambu Lab printer model (e.g. X1 Carbon)
- select Plate Type (e.g. Textured PEI Plate)

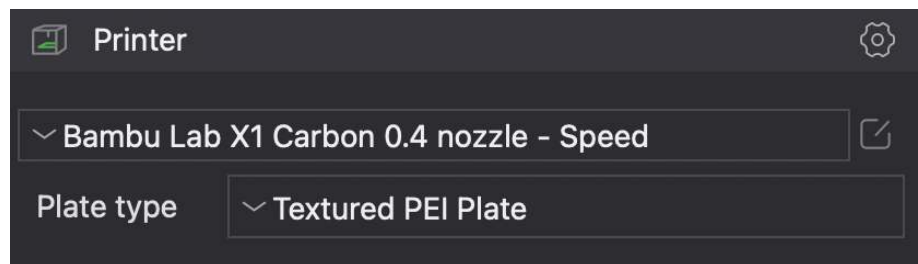


Figure 13: Printer and Plate selection

4. Select Filament Profile

Under **Prepare > Filament**

- select PLA type (e.g. Bambu PLA Basic)

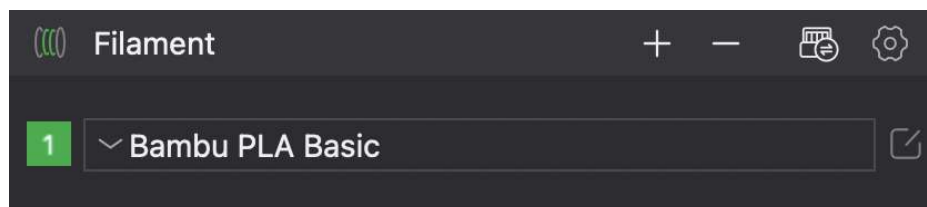


Figure 14: Filament selection

5. Enable Supports

Under **Prepare > Support**

- select Enable Support
- select (normal(auto))
- set threshold angle to 30 degrees.
- leave build plate unchecked
- select default for filament for supports.

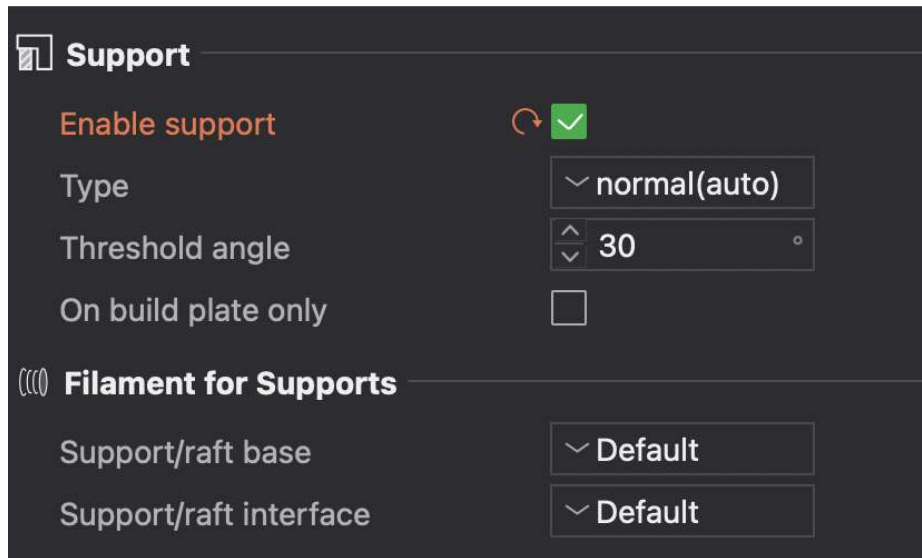


Figure 15: Support settings

6. Visualize in Preview

Select **Preview** in the upper left corner and visualize print process.

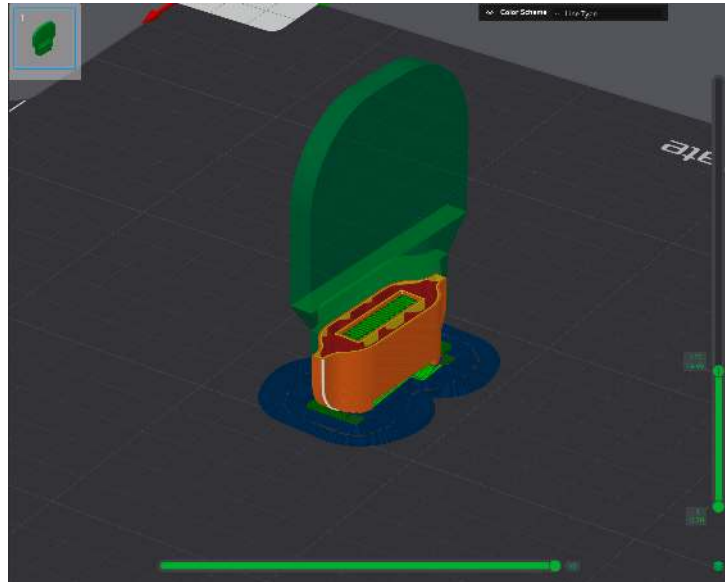


Figure 16: Visualize Print Process

7. Print Device

Select **Print Plate** in upper right corner

- Check print time and ensure pickup after given amount of time
- Select printer profile if more than one printer connected to Bambu Studio
- select bed leveling
- select flow dynamics calibration
- enable AMS

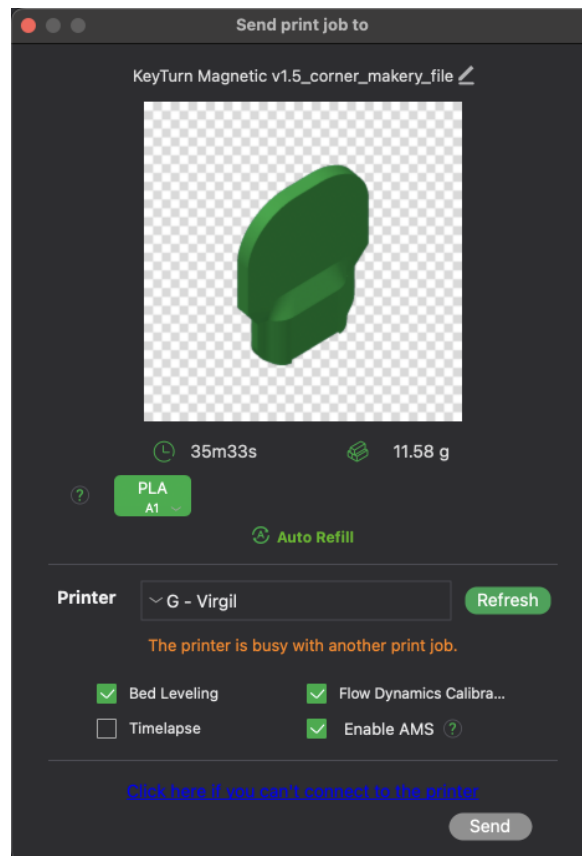


Figure 17: Setup printer profile

Section 2: Bambu Printer Setup

1. Load Filament

In the top shelf compartment, load the PLA filament roll of choice in and close.

2. Prepare Printing Bed

Ensure the build plate is clean and properly installed (no debris, plate aligned).

Section 3: During Printing Process

1. Monitor print start

Watch the first few layers to ensure good adhesion and no toppling of the device.

Section 4: Post Print Procedures

1. Remove Print

Wait 5 minutes after the printer stops running to ensure print is at a safe temperature.

2. Carefully remove supports

The supports are designed to be broken off by hand. If necessary, use small flush cutters.



Figure 18: Removal of supports

3. Sanding and Finishing (optional):

Sand rough edges if needed.

Instructions for Use

Initial Setup: Attaching the KeyTurn to the user

Step 1: Open the button connector (Figure 19) attached to the clear plastic on the Retraction Mechanism



Figure 19: Location of button connector on Retraction Mechanism

Step 2. Thread the clear plastic loop through the rectangular hole (Figure 20) at the top of the KeyTurn

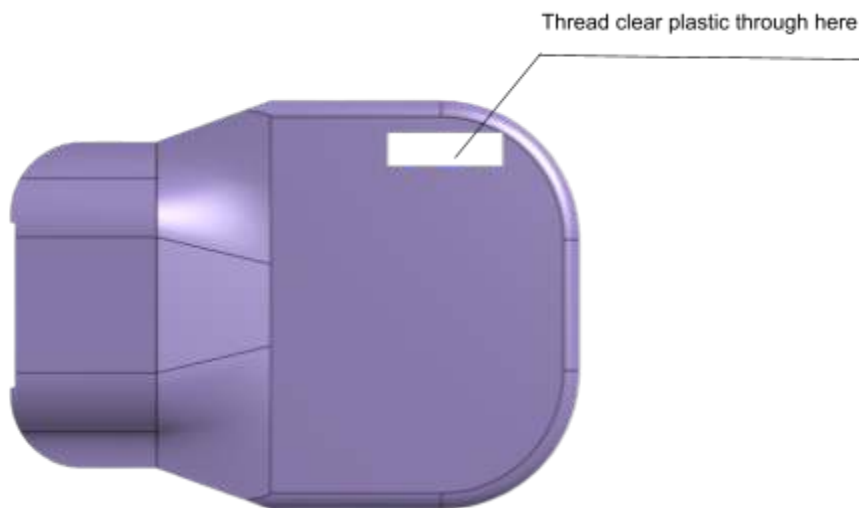


Figure 20: Location of rectangular hole on KeyTurn

Step 3. Close the button connector.

Step 4. Attach Retraction Mechanism using attachment methods (Figure 21) to preferred location on user or other object such as a backpack at waist height.

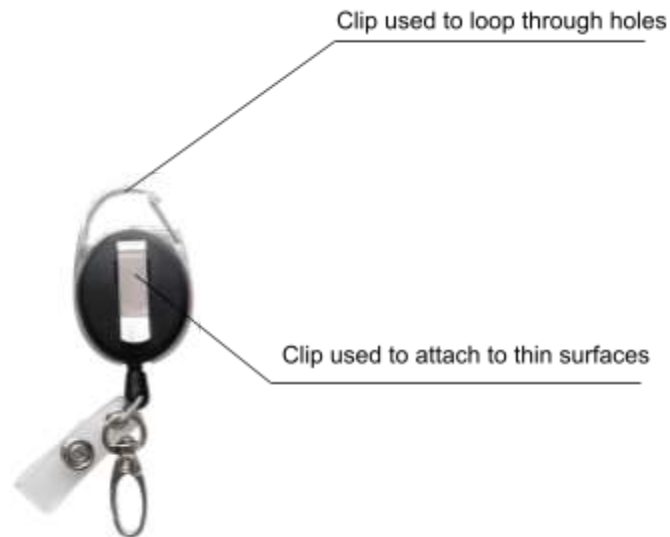


Figure 21: Location of attachment methods on Retraction Mechanism

Section 2. Using the KeyTurn

Step 1: Press the Key Turn onto the thumb-turn lock.

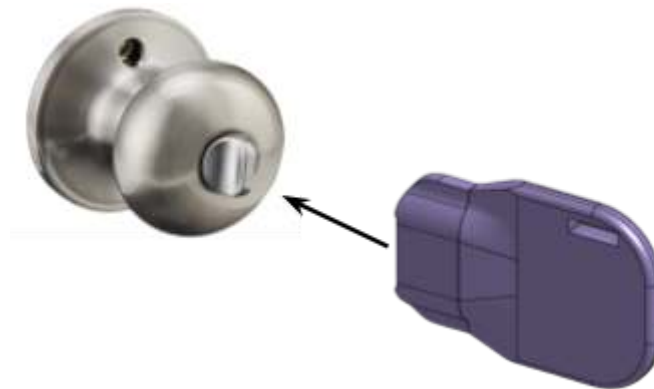


Figure 22: Diagram of inserting KeyTurn onto Thumb Turn Lock

Step 2. Turn the KeyTurn in the direction to unlock the door (direction based on personal doorknob design)

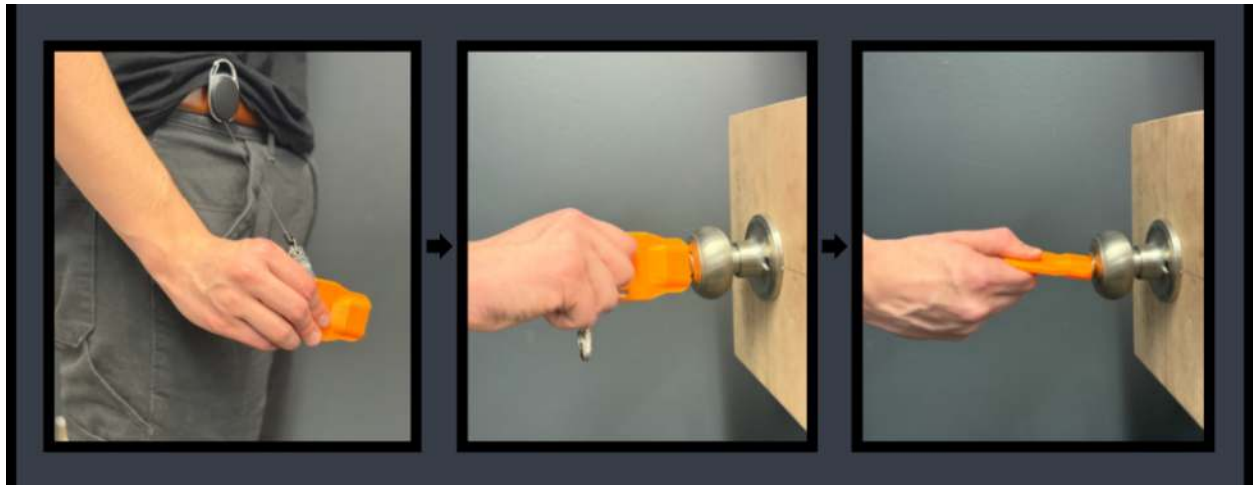


Figure 23: Demonstration of using the KeyTurn in 3 steps: (1) pulling the KeyTurn from the waist, (2) inserting the device onto the thumb turn lock and (3) turning the device.

Step 3. Turn the KeyTurn in the opposite direction to lock the door